



Intelligence

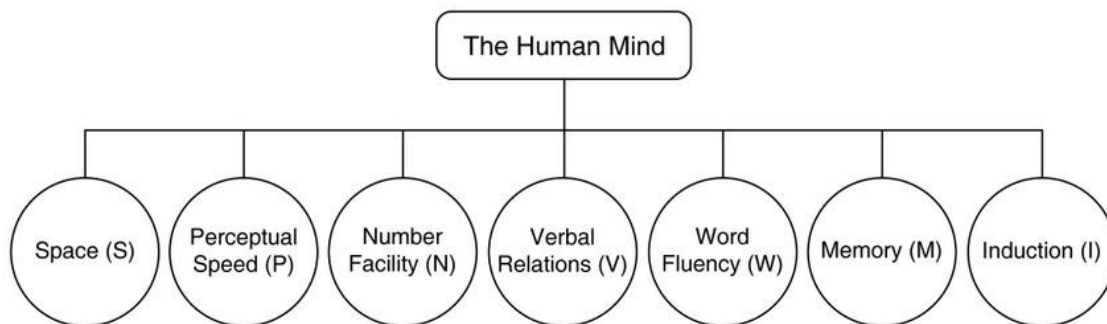
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RECAP

- Intelligence: The ability to learn from one's experiences, acquire knowledge, and use resources effectively in adapting to new situations or solving problems
- Binet's uni-factor theory: Mental age
- Spearman G factor: general intelligence
- Spearman s factor: specific intelligence

Theories of Intelligence: Thurstone's Primary Mental Abilities

- Thurstone's primary mental abilities or the group factor theory (1938)
- Proposed multiple distinct mental abilities or factors rather than the 'g' factor
- Seven primary mental abilities:** Relatively independent of each other
- Later studies: interconnection between the abilities (hinting on g factor)



(Thurstone, 1938)



LL Thurstone, an American. Of

Theories of Intelligence: Thurstone's Primary Mental Abilities (cont'd...)

Space (S)

Ability to mentally visualise and manipulate visual images and spaces
E.g., spatial reasoning and spatial relationships;
interior designer, graphic designer, aerospace engineering

Perceptual Speed (P)

Ability to quickly and accurately perceive visual details and discriminate between different stimuli
E.g., video game tester, radiologist, and quality control inspector

Number Facility (N)

Capacity to work with numbers, perform mathematical calculations
E.g., data scientist, engineer, accountant

Verbal Relations (V)

Ability to understand and use language effectively
E.g., reading comprehension, verbal reasoning, vocabulary

Word Fluency (W)

Ability to generate words and ideas quickly and fluently
E.g., spontaneous storytelling, improvisation

Memory (M)

Ability to encode, store, retrieve information, and make connections between ideas and form associations
E.g., academic researcher, medical professionals, librarian

Induction (I)

Ability to identify patterns, make generalisations, and draw conclusions based on incomplete information
E.g., problem-solving, critical thinking, inference-making

Theories of Intelligence: Thurstone's Primary Mental Abilities (cont'd...)

Criticism:

- Interdependence of Abilities: abilities often overlap or correlate, suggesting the influence of an underlying general intelligence factor (g)
- Oversimplification: Simplifies intelligence by focusing on seven factors only

Gardner's multiple intelligence

- Intelligence is not single entity but distinct modalities
- Selectively independent faculties
 - Linguistic Intelligence
 - Logical-mathematical Intelligence
 - Spatial Intelligence
 - Bodily Kinesthetics Intelligence (Body smart)
 - Musical Intelligence
 - Interpersonal intelligence
 - Intrapersonal intelligence (Self smart)
 - Naturalistic intelligence
 - Existential intelligence (possible 9th)

29/03/2026



Theories of Intelligence: Cattell's Theory of Intelligence

- Raymond Cattell (1963)
- *Crystallized intelligence*: acquired knowledge and skills; accumulated through life's experiences
- *Fluid intelligence*: information-processing capabilities, such as, reasoning and problem solving and adaptability in unfamiliar situations
- Fluid intelligence declines after adolescence, while crystallized intelligence grows through adulthood

Crystallized = Grows like a crystal
Fluid = Fades like flowing water

Crystallized intelligence examples:

- A chef who has memorized many recipes and cooking techniques over years of experience uses crystallized intelligence
- Playing trivia games where recalling facts and information is key

Fluid Intelligence examples:

- Figuring out how to fix a new type of gadget without instructions
- Using abstract reasoning to solve logic problems or pattern sequences on the spot

Theories of Intelligence: Cattell's Theory of Intelligence (cont'd...)

Criticism:

- The theory oversimplifies human intelligence by dividing it into only two types
- Fluid and crystallized intelligence are interrelated, not completely separate
- It focuses on intellectual abilities, ignoring creativity, emotional, and practical intelligence

Crystallized intelligence: 'I studied this for weeks!'

Fluid intelligence: 'Come on brain, figure this out in 5 minutes!'



Theories of Intelligence: Sternberg's Triarchic Theory

- Robert Sternberg (1988)
- Triarchic theory of intelligence
- Three kinds of intelligence
- **Analytical, creative, and practical intelligence**

Analytical intelligence

- Ability to break problems down into component parts, or analysis, for problem solving
- Measured by intelligence tests and academic achievement tests
- “Book smarts”

Practical intelligence

- Ability to use information to get along in life
- Tactful, manipulate situations to their advantage, use inside information to increase their odds of success
- “Street smarts”

Creative intelligence

- Ability to deal with new and different concepts
- Find new ways of solving problems (divergent thinking)

Theories of Intelligence: Sternberg's Triarchic Theory (cont'd...)

Example:

- Analytical: Understands theories and analyzes data
- Creative: Comes up with new research ideas or ways to explain concepts
- Practical: Manages time, organizes study groups, and applies knowledge in real life

Creativity

The process of solving problems by **combining ideas or behavior in new ways**

Convergent thinking

Type of thinking in which a problem is seen as having only one answer, and all lines of thinking will eventually lead to that single answer, using previous knowledge and logic

Examples:

- Solving a multiple-choice test where there is only one correct answer
- A detective using evidence to identify the single culprit in a crime, like Sherlock Holmes

Divergent thinking

Type of thinking in which a person starts from one point and comes up with many different ideas or possibilities based on that point

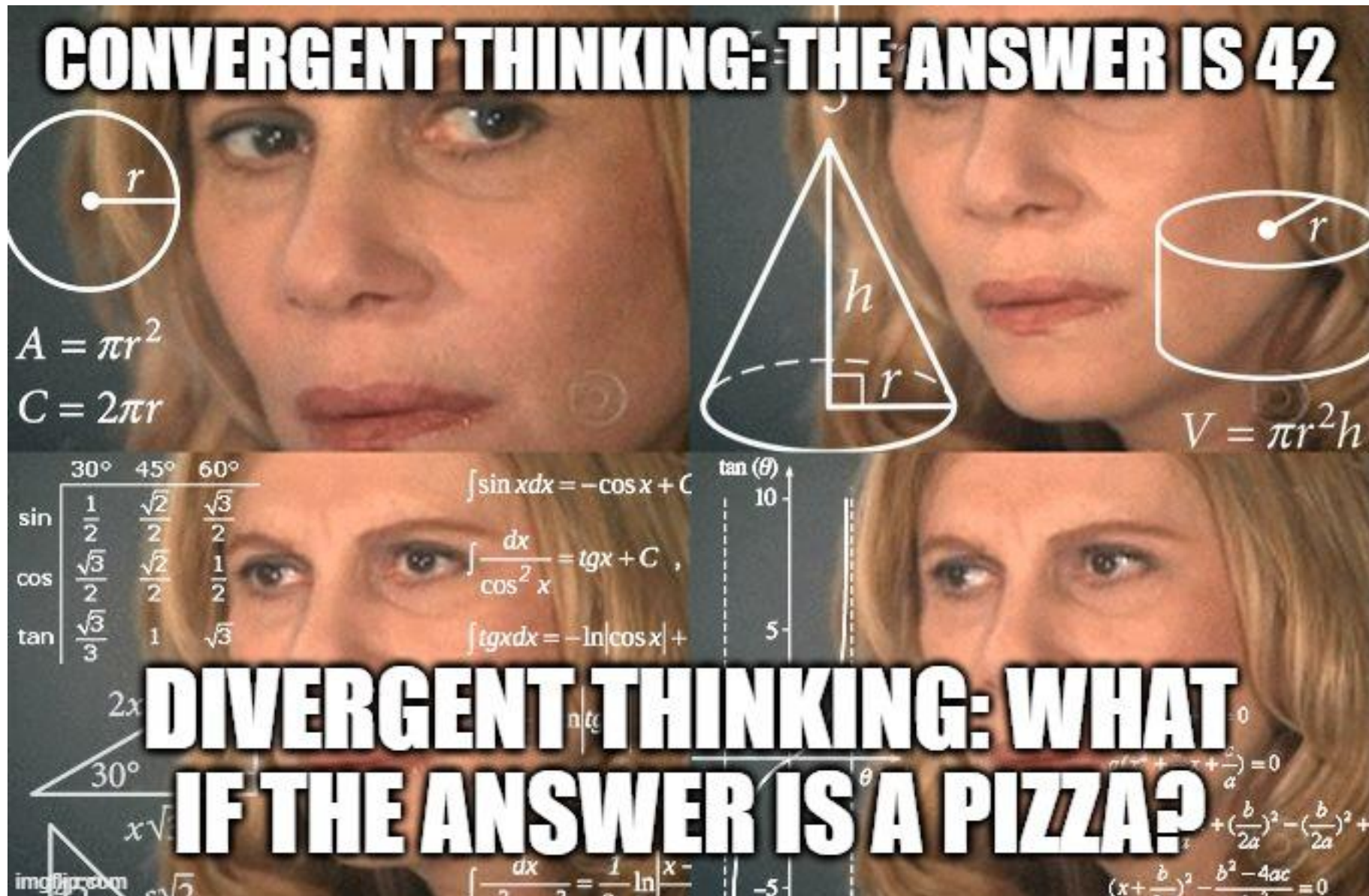
Examples:

- Brainstorming a variety of possible uses for a common object, like a paperclip
- Composing music or writing creative lyrics- producing numerous novel ideas without a single right answer

(Csikszentmihalyi, 1996)

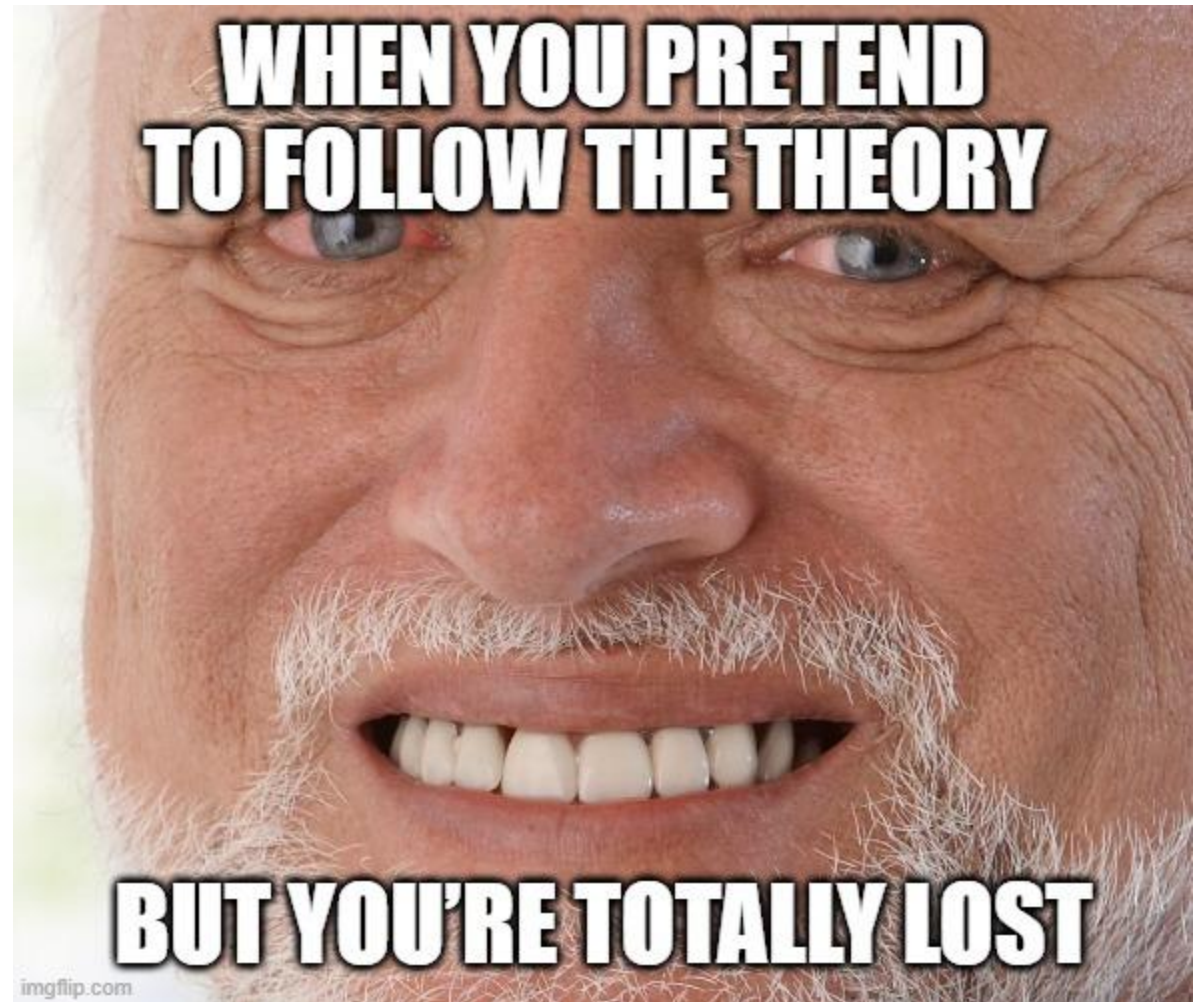
Divergent thinking = different and diverse kinds of responses

Convergent thinking = common sense kinds of responses



RECAP

- Thurstone Primary Mental Abilities
 - Seven Primary mental ability
- Gardner's multiple intelligence
 - Independent faculties
- Cattell's theory of intelligence
 - Crystallized intelligence
 - Fluid intelligence
- Sternberg's triarchic theory
 - Analytical
 - Practical
 - Creative intelligence
- Creativity
 - Convergent thinking
 - Divergent thinking



References

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- Feldman, R. S. (2009). *Essentials of understanding psychology* (8th ed.). McGraw-Hill.

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